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(Digital Postal Index Number)
DIGIPIN

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DIGIPIN (Digital Postal Index Number)

DIGIPIN is a **Digital Public Infrastructure (DPI)** and a standardized, geo-coded, interoperable addressing system in India established by the **Department of Posts, India**. This initiative seeks to provide simplified addressing solutions for seamless delivery of public and private services and to enable "**Address as a Service**" **(AaaS)** across the country.

Summary of sections:

1. **Overview & Launch Dates** (Updated to **May 27, 2025**).
2. **Core Concept & 10-Digit Architecture**.
3. **Bounding Box & Grid Levels** (4m x 4m units).
4. **Symbol Assignment Strategy** (Spiraling/Z-order).
5. **DHRUVA Ecosystem & Key Components**.
6. **Complete Abbreviation Table**.

Overview

DIGIPIN is an open-source national-level addressing grid developed by the **Department of Posts** in collaboration with **IIT Hyderabad** and **NRSC, ISRO**. It serves as a uniform address referencing framework available both **offline and online**.

The official Digital Postal Index Number system was launched on **May 27, 2025**.

📝 **LEGISLATIVE NOTE:** The official DIGIPIN system was formally launched on **May 27, 2025**, following a beta release on **July 19, 2024**, which remained open for public feedback until **September 22, 2024**. (Updated 2025).

Core Concept

DIGIPIN is visualized as an alphanumeric offline grid system that divides the geographical territory of India into uniform **4-meter by 4-meter (approx.)** units.

- Each of these **4m x 4m** units is assigned a unique **10-digit** alphanumeric code.
- The code is derived from the **latitude and longitude** coordinates of the unit.
- DIGIPIN solely represents a location and does not store any personal information, ensuring privacy.
- The system is designed to be permanent and does not change with updates to state, city, or locality names.

Code Architecture

The DIGIPIN is an encoding of the latitude and longitude of an address into a sequence of alphanumeric symbols using the following **16 symbols**: **2, 3, 4, 5, 6, 7, 8, 9, C, J, K, L, M, P, F, T**.

Partition Levels

The encoding is performed hierarchically across **10 levels**:

1. **Level-1 Partition:** A bounding box covering the entire country is split into **16 (4x4)** regions. Each region is labeled with one of the **16 symbols**, which becomes the first character of the code.
2. **Level-2 Partition:** Each level-1 region is subdivided into **16** subregions, each labeled with the same set of symbols. The second character identifies the specific subregion.
3. **Subsequent Levels:** This process is repeated for a total of **10 levels**.

The final **10-symbol** code uniquely identifies one of the **16^{10}** cells within the national bounding box.

Bounding Box and Grid Specifications

The bounding box used for the DIGIPIN system covers:

- **Longitude: 63.5 - 99.5 degrees East**
- **Latitude: 2.5 - 38.5 degrees North**

This area includes the maritime **Exclusive Economic Zone (EEZ)**, allowing DIGIPIN to provide addresses for Indian assets at sea, such as oil rigs. The maritime EEZ is computed based on a **200 nautical miles** extent from the coastline.

Grid Sizes at Various Levels

Code Length	Grid Width	Approx. Distance
1	9′	1000 km
2	2.25′	250 km
3	33.75′	62.5 km
4	8.44′	15.6 km
5	2.11′	3.9 km
6	0.53′	1 km
7	0.13′	250 m
8	0.03′	60 m
9	0.5′	15 m
10	0.12′	3.8 m

A code length of **6** represents an approximately **1km x 1km** region, while a full **10-digit** code precisely locates a **3.8m x 3.8m** unit.

Symbol Assignment Strategy

The symbols within each grid level are assigned using a **Z-order** or **Peano curve** logic, but with a specific **spiraling pattern** that begins from the **bottom-left (South-West)** corner of the country.

- The grid numbering starts with the symbol **2** at the bottom-left corner of the grid.
- The assignment follows a specific sequence of the **16 symbols** across the cells.
- This pattern ensures that nearby locations share similar prefixes in their DIGIPIN, which is essential for efficient spatial indexing and logistics.

DHRUVA (Digital Hub for Reference & Unique Virtual Address)

DHRUVA is the comprehensive digital address ecosystem where DIGIPIN acts as the foundational layer. It is designed to function as a **Digital Public Infrastructure (DPI)** for addressing, similar to how **UPI** functions for payments.

Core Objectives of DHRUVA

1. **Standardization:** Establishing a uniform addressing format across India.
2. **Interoperability:** Ensuring that digital addresses can be used seamlessly across different platforms (e-commerce, emergency services, logistics).
3. **Privacy by Design:** Decoupling physical locations from personal identity.
4. **Inclusivity:** Providing a formal address to every citizen, including those in unplanned settlements or remote areas.

Key Components of the Ecosystem

- **Central Mapper (CM):** The primary authority managing the translation between DIGIPIN and other address formats.

- **Authorized Address Validation Agency (AAVA):** Entities authorized to verify the authenticity of a physical address associated with a DIGIPIN.
- **Address Information Provider (AIP):** Entities that provide or host address data.
- **Address Information User (AIU):** Services (like delivery apps or government portals) that utilize the DIGIPIN for service fulfillment.

User Interfaces & Features

The DHRUVA ecosystem provides several key interfaces for users:

- **Digital Address Creation:** Users can generate a unique digital address for their specific location.
- **Consent-based Data Sharing:** Residents have control over who accesses their precise location data.
- **Secure Address Verification:** Prevents fraud and ensures accurate service delivery.
- **Virtual Address:** Users can create a handle (e.g., dakguru@dhruva) that masks their technical DIGIPIN or physical coordinates, just like a **UPI ID**.

Benefits of DIGIPIN and DHRUVA

- **Disaster Management:** Precise location identification for rescue operations in areas where landmarks may have been destroyed.
- **Logistics Efficiency:** Reducing "last-mile" delivery failures caused by vague or incorrect manual addresses.
- **Governance:** Better planning for urban utilities, health services (integrated with **Ayushman Bharat Digital Mission - ABDM**), and emergency response (Police/Ambulance).

- **Economic Impact:** Aligning with the **National Geospatial Policy 2022** to strengthen the information economy.

Important Abbreviations

Abbreviation	Full Form
AaaS	Address as a Service
AAVA	Authorized Address Validation Agency
ABDM	Ayushman Bharat Digital Mission
AIA	Address Information Agent
AIP	Address Information Provider
AIU	Address Information User
API	Application Programming Interface
CM	Central Mapper
DHRUVA	Digital Hub for Reference & Unique Virtual Address
DIGIPIN	Digital Postal Index Number
DPDPA	Digital Personal Data Protection Act, 2023
DPI	Digital Public Infrastructure
GIS	Geographic Information System
GNSS	Global Navigation Satellite System
KYC	Know Your Customer
NHA	National Health Authority
NPCI	National Payments Corporation of India
ONDC	Open Network for Digital Commerce
PIN	Postal Index Number
SDG	Sustainable Development Goal
UIDAI	Unique Identification Authority of India
UPI	Unified Payments Interface
UPRN	Unique Property Reference Number
UPU	Universal Postal Union